

ORIGINAL ARTICLE

Fibroblast proliferative capacity predicts prognosis in idiopathic pulmonary fibrosis and identifies PITX1 as a growth-associated gene

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ABSTRACT

Background: Idiopathic pulmonary fibrosis (IPF) is a progressive fibrotic lung disease with poor prognosis. Lung fibroblasts isolated from patients with IPF exhibit enhanced proliferative capacity; however, the molecular determinants linking fibroblast proliferation to clinical outcomes remain incompletely understood.

Methods: Fibroblast growth slopes were measured in primary lung fibroblast derived from patients with IPF and control subjects. Transcriptome microarray data were integrated with fibroblast growth kinetics to identify growth slope-associated genes. Differential expression analysis was performed between IPF and control fibroblasts, and candidate genes were further evaluated at the mRNA and protein levels in lung fibroblasts and bronchoalveolar lavage (BAL) fluids. Associations with clinical outcomes were assessed.

Results: Fibroblasts derived from patients with IPF showed significantly higher growth slope than control fibroblasts, and increased fibroblast growth slope were associated with reduced survival in IPF. Integration of fibroblast growth kinetics with transcriptomic analysis identified a fibroblast growth slope-associated gene signature, from which PITX1 emerged as a disease-relevant candidate gene. PITX1 mRNA and protein levels were elevated in IPF fibroblasts compared with controls. In BAL fluids, PITX1 protein levels were significantly higher in patients with IPF and were associated with clinical outcomes, although they did not correlate with pulmonary function parameters.



Received: 22 April 2026 | Accepted: 14 June 2026

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Conclusions: Enhanced fibroblast proliferative capacity is associated with poor prognosis in IPF, and PITX1 represents a fibroblast growth slope-associated gene reflecting fibroblast heterogeneity and prognostic relevance. These findings suggest that assessment of fibroblast growth behavior and PITX1 expression may provide insight into disease progression in IPF, although further validation studies are required.

Key words: idiopathic pulmonary fibrosis, fibroblast proliferation, PITX1, prognosis

Introduction

Idiopathic pulmonary fibrosis (IPF) is a chronically progressive fibrotic interstitial lung disease characterized by excessive accumulation of myofibroblasts and extracellular matrix within the lung parenchyma, leading to irreversible architectural distortion and decline in lung function (1). Disease progression is associated with increasing lung stiffness and reduction in lung volumes, which correlate with long-term mortality in patients with IPF (2). In normal wound healing, myofibroblasts that are transiently activated during tissue repair undergo apoptosis once re-epithelialization is achieved, allowing resolution of extracellular matrix deposition (3). In contrast, IPF is thought to result from aberrant wound healing in response to repetitive epithelial injury, leading to persistent fibroblast activation, excessive extracellular matrix accumulation, and formation of fibroblastic foci (1). The extent of fibroblastic foci in lung tissue has been shown to inversely correlate with patient survival, underscoring the clinical relevance of fibroblast behavior in IPF pathogenesis (4,5).

Fibroblast accumulation in IPF reflects both microenvironmental cues and intrinsic cellular alterations. Profibrotic mediators released following epithelial injury promote fibroblast proliferation and activation, while IPF-derived fibroblasts exhibit distinct phenotypic features, including enhanced extracellular matrix production and altered survival signaling, compared with fibroblasts from normal lungs (6-8). These observations suggest that intrinsic differences in fibroblast behavior may contribute to disease progression and clinical heterogeneity in IPF.

To date, several transcriptomic studies have compared gene expression profiles between IPF and normal lung fibroblasts using bulk RNA microarrays or, more recently, single-cell RNA sequencing approaches (9-13). These studies have provided valuable insights into fibroblast activation states and lineage trajectories in IPF. However, relatively few studies have directly examined how fibroblast transcriptomic profiles relate to functional cellular phenotypes, such as proliferative capacity, that may influence disease progression.

Previous *in vitro* studies have demonstrated that fibroblasts derived from fibrotic lungs exhibit resistance to apoptosis and altered responses to cytokine signaling, including IL-6-mediated survival pathways (14-16). These findings led us to hypothesize that variability in fibroblast proliferative behavior may be associated with clinical outcomes in IPF. In this study, we therefore assessed the relationship between *in vitro* fibroblast growth slope and patient survival and integrated fibroblast growth kinetics with genome-wide transcriptomic analysis to identify genes associated with fibroblast proliferation and prognosis in IPF.

Materials and Methods

Study subjects

Lung Fibroblasts, tissues and BAL fluids from patients with IPF were obtained from the biobank of Soonchunhyang University Hospital, Bucheon, Korea after approval of the study protocol by the Ethics Committee of Soonchunhyang University Hospital (SCHCA-IRB-2018-10-034 and 201910-BR-058).

Informed written consents for study participation and sample donation were obtained from each subject. All subjects were examined by physicians to obtain their medical history and undergo chest X-ray, high-resolution chest computed tomography (HRCT) and pulmonary function tests. None of the IPF patients had any evidence of underlying collagen vascular diseases according to laboratory tests and clinical symptoms. The diagnosis of IPF was based on the 2011 (17) and the 2018 guidelines proposed by ATS/ERS/JRS/ALAT Committee (18). None of the patients diagnosed with IPF exhibited any signs of underlying collagen vascular diseases, as confirmed by laboratory tests and clinical symptoms. Controls did not show any respiratory symptoms, as confirmed by a screening questionnaire, and had forced expiratory volume in 1 s (FEV1) and forced vital capacity (FVC) levels above 80%, along with normal chest radiographs.

Fibroblast culture and proliferation assay

Lung tissue fibroblasts were obtained by surgical biopsy in 33 patients with IPF (histologically confirmed UIP) and from 10 histologically normal lung distant from resected tumor (stage 1 or 2), and then cultured as described in our previous study (19,20). Briefly, lung specimens were finely minced and placed into 150 cm² cell culture flasks with tissue culture media (TCM) consisting of DMEM (Lonza Walkersville, Inc., MD, USA), 10% fetal bovine serum (Thermo Fisher Scientific Inc., Rockford, IL, USA), 2mmol/L glutamine, and 1% penicillin-streptomycin-amphotericin (Lonza Walkersville, Inc). Cells were maintained at 37°C in a 5% CO₂ incubator and serially sub-cultured to yield a morphologically homogeneous population of adherent fibroblasts under the microscopy and α -smooth muscle actin (SMA) immunohistochemical stain (Abcam, Cambridge, MA, USA) until the fourth passage. Fifth-passage fibroblasts (2.5×10^6) were seeded in 1mL TCM in 10cm² dishes. After reaching 80% confluence, the fibroblasts were washed twice with DPBS (Dulbecco's Phosphate-Buffered Saline, Thermo Fisher Scientific Inc.) and used for separation of total RNA and proteins. The proliferation rate was measured using the reagent MTT (3-[4,5-dimethylthiazol-2-yl]-2,5

diphenyl tetrazolium bromide, Sigma, St. Louis, MO, USA) assay as described (21). Briefly, the 5th passage - fibroblasts were plated in 96-well culture plates (2.5×10^3 cells/well) and incubated at 37°C in 5% CO₂ /95% air in TCM supplemented with 10% FBS. At 0, 24, 48, and 72 hours of incubation, the medium was replaced using serum free media with MTT (5ug/ul) according to manufacturer's instructions. After incubation for a further 5 hours, a solubilizing solution was added to dissolve the insoluble purple formazan product into a colored solution for overnight. The absorbance was measured at a wavelength of 550 nm using a reference wavelength of 690 nm using a microplate reader (Molecular Devices, LLC, USA). Fibroblast growth slopes were generated by the plotting of optical density (OD) over time from day 0. For each fibroblast growth slope, the best-fit lines were calculated using regression analysis, and the slopes of those lines were used as estimates of the fibroblast growth slope of each cell population per day.

Correlation of fibroblast proliferation rates with gene expression intensities of transcriptome microarray

The transcriptome microarray data of the eight IPF-fibroblasts and four control-fibroblasts, which covered 27,455 curated and putative genes, were obtained from the Gene Expression Omnibus (GEO) database under the accession number GSE71351 that was previously published (19). Spearman's rank correlation coefficient was applied to evaluate correlation between the expression intensities of the genes and fibroblast growth slope. Fold change of gene expression was calculated as follows: the mean of expression levels of IPF-fibroblasts divided by that of control fibroblasts if the levels were higher in the IPF than the controls. In cases of countertrend, fold change was determined by dividing the mean of control group by that of the IPF group and presented as minus value. Significance was defined if between groups difference of gene expression intensities had *p*-value of less than 0.05 by t-test and absolute fold change of greater than 2, as previously described (19). Gene functional annotation was performed by the WebGestalt (<http://www.webgestalt.org/>) (22), which was based on hypergeometric

distribution to show the overrepresented gene ontology categories ($p < 0.05$) (23). Genes were classified by biological process, cellular component and molecular function via the GO (Gene Ontology) and KEGG (Kyoto Encyclopedia of Genes and Genomes) databases.

Validation of PITX1 mRNA expressed by the cultured fibroblasts

Total RNA was extracted using TRI reagent (Ambion, Carlsbad, CA, USA). The cell pellets were prepared in RIPA buffer for immunoblot analysis, and protein concentrations were measured using a BCA kit (Thermo Fisher Scientific Inc.). For RT-PCT and real time PCR, total RNA was extracted from the fibroblasts and converted to cDNA, which was amplified and purified using an Illumina® Total Prep™ RNA Amplification Kit (Ambion, Carlsbad, CA, USA). The quality and quantity of the extracted RNA were examined by a RNA quantification reagent (Ribogreen®, Invitrogen, Carlsbad, CA, USA). Total RNA (1 µg) was suspended in diethylpyrocarbonate-treated water with 0.5 µg 10 mM dNTPs and oligodeoxythymidine, heated at 65°C for 5 min, and then cooled on ice. PCR amplification was performed for 30 cycles (5 min at 94°C, 30 s at 94°C, 30 s at 60°C, and 30 s at 72°C) with a final extension at 72°C for 7 min. The PCR mixture (20 µL) contained 1 µg cDNA, 10 µL 2 × Power SYBR Green PCR Master Mix (Applied Biosystems, Foster City, CA, USA), and 1 µL 10 pmol forward and reverse primers. The following primer sequences were used: PITX1 (sense) 5'-CACTCCATGTCCCCATGCTC-3' and (anti-sense) 5'-CCGCTACAGCATCGTCTCTT-3'; β-actin (sense) 5'-GGACTTCGAGCAAGAGATGG-3' and (antisense) 5'-AGCACTGTGTTGGCGTACAG-3'. The PCR products were separated on 1.0% agarose gel containing ethidium bromide in Tris-borate EDTA buffer at 100 V for 40 min and visualized under ultraviolet light. The PITX1 band intensities were normalized to those of β-actin. Real-time PCR was executed using the StepOne™ Real-Time PCR System (Applied Biosystems) in a two-step procedure: denaturation at 95°C for 15 s and 60°C for 1 min, followed by melting at 95°C for 15 s, 60°C for

1 min, and 95°C for 15 s. Expression levels were calculated using the 2-ΔΔCT method (24) and expressed as the relative fold change after normalization to PPIA.

Measurement of PITX1 protein in the lysates of fibroblasts and in BAL fluids using western blot and enzyme-linked immunosorbent assay

Primary lung fibroblast cells lysed in RIPA buffer (Sigma-Aldrich, St. Louis, MO, USA) with protease-inhibitor (Thermo Fisher) containing proteinase and phosphatase inhibitor cocktails (Roche Diagnostics, Basel, Switzerland) and PITX1 protein were analyzed using western blot. Total protein concentrations were measured using a BCA kit (Thermo Fisher). For each experiment, equal quantities of total protein were resolved using 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis. The proteins were transferred onto polyvinylidene difluoride membranes (Millipore, Billerica, MA, USA). The immunoblots were subsequently blocked in 5% skimmed milk and incubated for 24 h at 4°C with the following primary antibodies: anti-polyclonal PITX1 (1:500; OriGene, Rockville, MD, USA) and anti-β-actin (1:50000; Sigma-Aldrich). After washing several times with phosphate-buffered saline containing Tween 20, the membranes were incubated with an anti-rabbit or an anti-mouse immunoglobulin G (IgG) horseradish peroxidase-conjugated secondary antibody (Gendepot, Katy, TX, USA). The membranes were analyzed by chemiluminescence (Thermo Fisher Scientific and Bio-Rad, Berkeley, CA, USA) using the ChemiDoc™ Touch Imaging System (Bio-Rad). PITX1 protein concentrations in BAL fluids were measured in normal controls and IPF patients using an ELISA kit (MyBioSource, San Diego, CA, USA) according to the manufacturer's recommendation. The lower limit of detection was 0.094 ng/mL; values below this limit were regarded as 0 ng/mL. The coefficients of variance for inter- and intra-assays were less than 15%.

Statistical analysis

Statistical analyses were performed using the Biomedical Analysis Workspace (hyper.schedulerju.com). The normality of the distribution was tested

using the Shapiro–Wilk test. Student’s *t* test or the Mann–Whitney U test was used to analyze continuous data. Categorical data were compared using the χ^2 test. Spearman’s rank correlation coefficient was applied to evaluate correlation between the gene expression intensities and fibroblast growth slopes of the IPF groups. A receiver operating characteristic (ROC) curve, AUC, and cut-off value were calculated using MedCalc (MedCalc Software v. 12.2.1.0, Ostend, Belgium) (25). The optimal cutoff level of PITX1 was calculated using Cutoff Finder (26). Survival rates were analyzed by Kaplan–Meier’s method and compared between the groups using log-rank test. Values of $p < 0.05$ were considered significant. Skewed data are presented as median value with the 25th and 75th percentile, and normally distributed data are expressed as mean $\pm$ standard error of the mean. *p*-values < 0.05 were considered significant.

Results

Clinical characteristics of the study groups

Lung fibroblasts were cultured from 33 IPF-lungs and 10 control lungs. The patients with IPF had significantly lower FVC and DL_{CO} values compared with those of control ($p < 0.05$) (Table 1). The fibroblast growth slope was significantly increased in IPF more

than control [(0.79 $\pm$ 0.4) vs. (0.5 $\pm$ 0.16) / 3days, $p = 0.002$, Figure 1]. In IPF group, the slope of fibroblast growth slope was analyzed in 33 subjects followed up for 1 to 10 years with respect to the survival rate. When the subjects were divided into two groups with a cut-off value of 1.018 / 3days, the survival rate was significantly lower in the group greater than that ($n = 11$) compared with that in the group of less than that ($n = 22$) (hazard ratio = 3.22 CI: 1–10.38; $p = 0.04$, Figure 2). When compared between the two groups, there were no differences in clinical and physiological parameters (Supplemental Table 1).

Fibroblast growth slope–associated gene signature in IPF fibroblasts

To identify transcriptional programs associated with fibroblast proliferative capacity in IPF, correlations between gene expression levels and fibroblast growth slopes were examined in samples for which both fibroblast growth slope measurements and transcriptome microarray data were available. Among a total of 33 fibroblast cultures with fibroblast growth slope measurements, transcriptome data from the GEO database (GSE71351) could be matched to eight IPF-derived fibroblast samples. Transcriptome data from control fibroblasts ($n = 4$) from the same dataset were included for comparative differential expression analyses. After quality control and filtering, 15,020 genes

Table 1. Clinical characteristics of the study subjects.

Variables	Control	IPF	p-value
No.	10	33	–
Age (years)	69(63–76)	75(71–80)	0.147
Sex (male/female)	4/6	15/18	0.761
Smokers (NS/ES/SM)	7/1/2	16/6/11	0.578
Smoke_amount (pack year)	7 $\pm$ 14.94	14.7 $\pm$ 19.5	0.249
FVC (% pred.)	90 $\pm$ 18	75 $\pm$ 17.62	0.033
DLCO (% pred.)	90 $\pm$ 17	67 $\pm$ 17	0.001
Death/Survival (No.)	2/8	13/20	0.26
Fibroblast growth slope (%/day)	0.5 $\pm$ 0.16	0.79 $\pm$ 0.4	0.002

Abbreviations: IPF, idiopathic pulmonary fibrosis; NS/ES/SM, non-smokers/ex-smokers/smokers; FVC, forced vital capacity; DLCO, diffusion capacity of the lung for carbon monoxide. Data with skewed distributions are presented as medians (25th–75th percentiles), and comparisons between two groups were performed using the Mann–Whitney U test. Normally distributed data are presented as means $\pm$ standard errors of the mean. Comparisons of the variables between the groups were performed using Mann–Whitney U test. *p*-values < 0.05 were considered statistically significant.

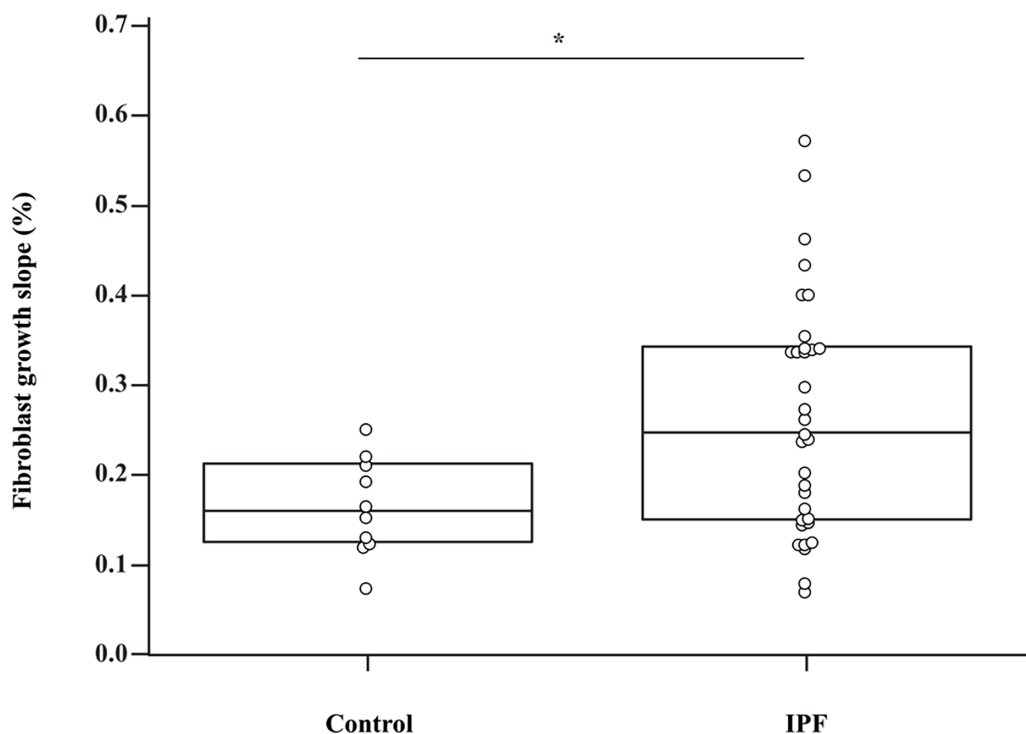


Figure 1. Comparison of fibroblast growth slope between controls and patients with IPF. Fibroblast growth slope (%) of fibroblasts derived from controls ($n = 10$) and patients with IPF ($n = 33$) are presented as box plots. The fibroblast growth slope in the IPF was significantly higher than that in the control ($*p = 0.002$). Each circle represents an individual sample, and the horizontal line within the box indicates the median, with the whiskers representing the interquartile range (25th to 75th percentiles).

were retained for analysis, among which 416 genes showed a significant correlation with fibroblast growth slopes of IPF-derived fibroblasts (Spearman correlation, $p < 0.05$) (Figure 3, Supplemental Table 2). These genes were defined as fibroblast growth slope-associated genes and were subjected to subsequent analyses. GO enrichment analysis indicated that the fibroblast growth slope-associated genes were primarily involved in biological regulation and metabolic processes and were predominantly localized to the membrane and nucleus (Figure 4A). More detailed enrichment analysis demonstrated that these genes were strongly enriched in biological processes related to connective tissue replacement and translational activity, including cytoplasmic translation and ribosome-associated terms. Consistent with these findings, cellular component analysis highlighted enrichment of ribosomal structures, while molecular function analysis revealed overrepresentation of regulatory activities

such as GTPase inhibitor activity and protein binding (Figure 4B, Supplemental Table 3).

To identify disease-relevant regulators linking this growth-associated transcriptional program to IPF-specific gene expression changes, the 416-fibroblast growth slope-associated genes were intersected with genes differentially expressed between patients with IPF and controls using transcriptome data from the same GEO dataset (GSE71351). Differential expression analysis based on a generalized linear model (GLM) identified 1,274 differentially expressed genes between the two groups ($p < 0.05$) (Supplemental Table 4). Among these, 33 genes were also correlated with fibroblast growth slope, and 7 genes exhibited more than a two-fold difference in expression between IPF and control samples (Table 2). To prioritize candidate genes for further validation, we considered the magnitude of differential expression, association with fibroblast growth slope, annotation reliability,

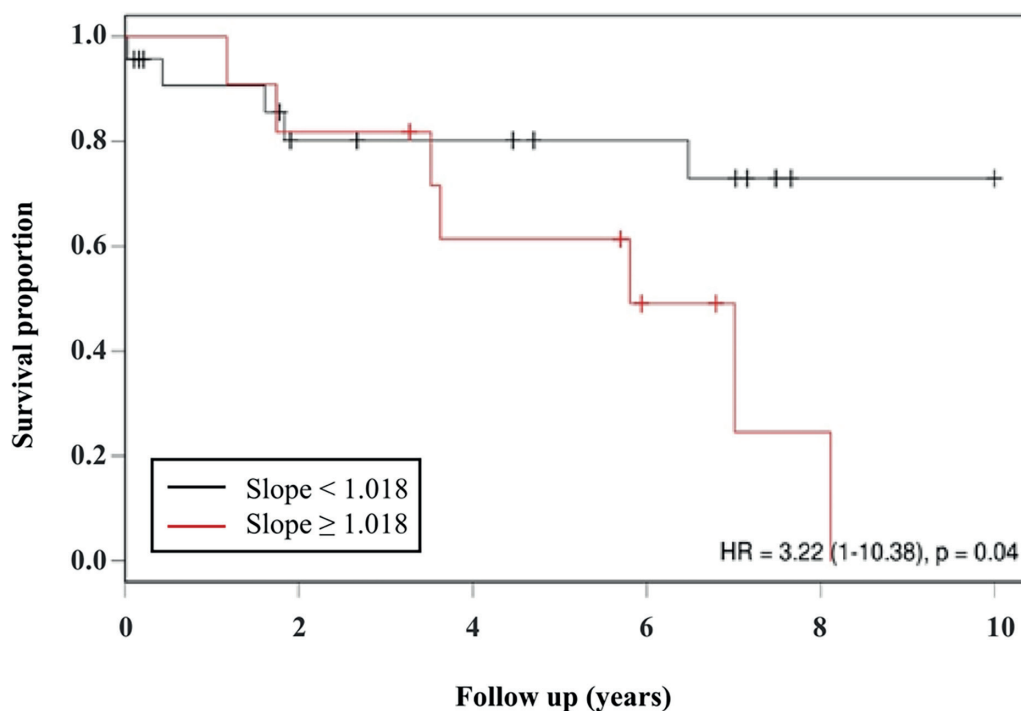


Figure 2. Kaplan–Meier survival curves of 33 subjects with IPF based on fibroblast growth slope. Patients were followed up for a period of 1 to 10 years. Survival analysis was significantly lower in the group with a fibroblast growth slope ≥ 1.018 ($n = 11$, red line) compared to the group with a slope < 1.018 ($n = 22$, black line). The hazard ratio (HR) was 3.22 (95% CI, 1.00–10.38; $p = 0.04$; log-rank test). Tick marks on the curves indicate censored data. *Abbreviations:* HR, hazard ratio.

expression consistency, biological relevance, and feasibility of validation at the mRNA and protein levels. Among these candidate genes, *PITX1* showed the greatest fold change in transcriptomic expression. Two genes (*LOC100129906* and *LOC100134466*) were excluded from further analysis because their records were withdrawn following annotation updates. The remaining genes displayed markedly skewed expression distributions across samples, limiting their robustness as representative markers. Based on its strong association with fibroblast growth slopes, disease-specific differential expression, and biological relevance within the growth-associated functional landscape, *PITX1* was selected as the primary candidate gene. Subsequent analyses were therefore performed to quantitatively assess *PITX1* expression at both the mRNA and protein levels in lung fibroblasts, as well as to evaluate its protein levels in BALFs and their clinical implications in patients with IPF.

Quantitation of *PITX1* mRNA and protein levels in fibroblasts

Fibroblasts isolated from lungs of patients with IPF ($n = 14$) and control subjects ($n = 10$) were analyzed. *PITX1* mRNA expression levels, normalized to β -actin, were approximately two-fold higher in IPF fibroblasts than in control fibroblasts, as determined by reverse transcription PCR and quantitative real-time PCR ($p < 0.05$ and $p < 0.005$, respectively; Figure 5A–C). Consistently, *PITX1* protein levels normalized to β -actin were also increased approximately two-fold in IPF fibroblasts compared with control fibroblasts, as assessed by Western blot analysis ($p < 0.05$; Figure 5D and E).

PITX1 protein levels in BAL fluids and clinical implications

BAL fluids samples from patients with IPF ($n = 80$) and control subjects ($n = 30$) were analyzed, and

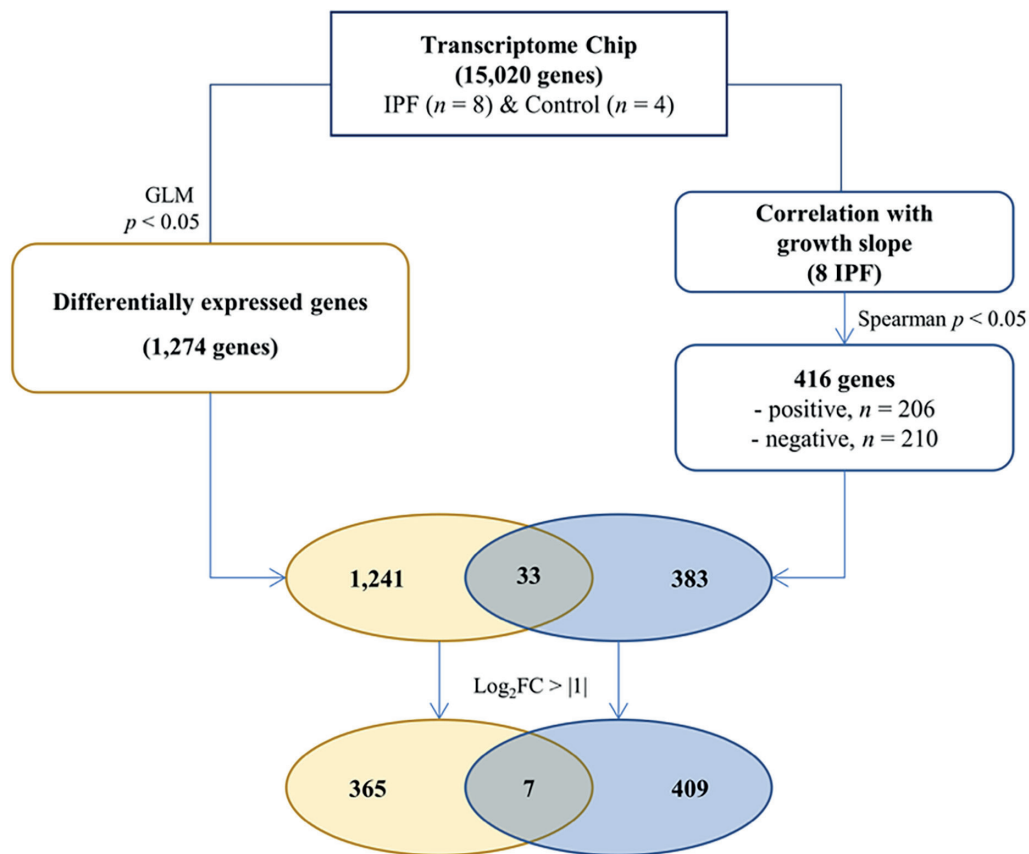


Figure 3. Flowchart of transcriptome analysis to identify genes associated with IPF and fibroblast growth slope. Transcriptome profiling (GSE71351) was performed using a chip containing 15,020 genes with samples from eight IPF patients and four controls. DEGs were identified between the IPF and controls using a GLM ($p < 0.05$), resulting in 1,274 genes. Simultaneously, Spearman correlation analysis was conducted to identify genes significantly associated with the fibroblast growth slope within the IPF group ($p < 0.05$), yielding 416 genes. The intersection of these two analyses identified 33 common genes. After applying $|\text{FC}| > 2$, seven candidate genes were finally selected as key factors potentially driving both the disease state and the aggressive cellular phenotype. Abbreviations: IPF, idiopathic pulmonary fibrosis; GLM, Generalized Linear Model; DEG, differentially expressed gene; FC, fold change.

the clinical characteristics of the study population are summarized in Table 3. PITX1 protein levels in BAL fluids, measured by ELISA, were significantly higher in patients with IPF than in controls [median (range): 0.61 (0–3.73) vs. 0 (0–0) ng/mL; $p < 0.001$] (Figure 6A). Receiver operating characteristic curve analysis identified a PITX1 cut-off value of 0.53 ng/mL, which discriminated patients with IPF from controls with a specificity of 100% and a sensitivity of 46.4% ($p < 0.0001$; Figure 6B). Among patients with IPF who were followed for 1–5 years ($n = 80$), Kaplan–Meier survival analysis demonstrated that patients

with higher BAL PITX1 levels (≥ 2.599 ng/mL, $n = 26$) had significantly better survival outcomes than those with lower PITX1 levels (< 2.599 ng/mL, $n = 54$) (hazard ratio, 0.29; 95% confidence interval, 0.102–0.821; $p = 0.02$; Figure 6C).

Discussion

In this study, we demonstrate that lung fibroblasts derived from patients with IPF exhibit enhanced proliferative capacity compared with control fibroblasts, and

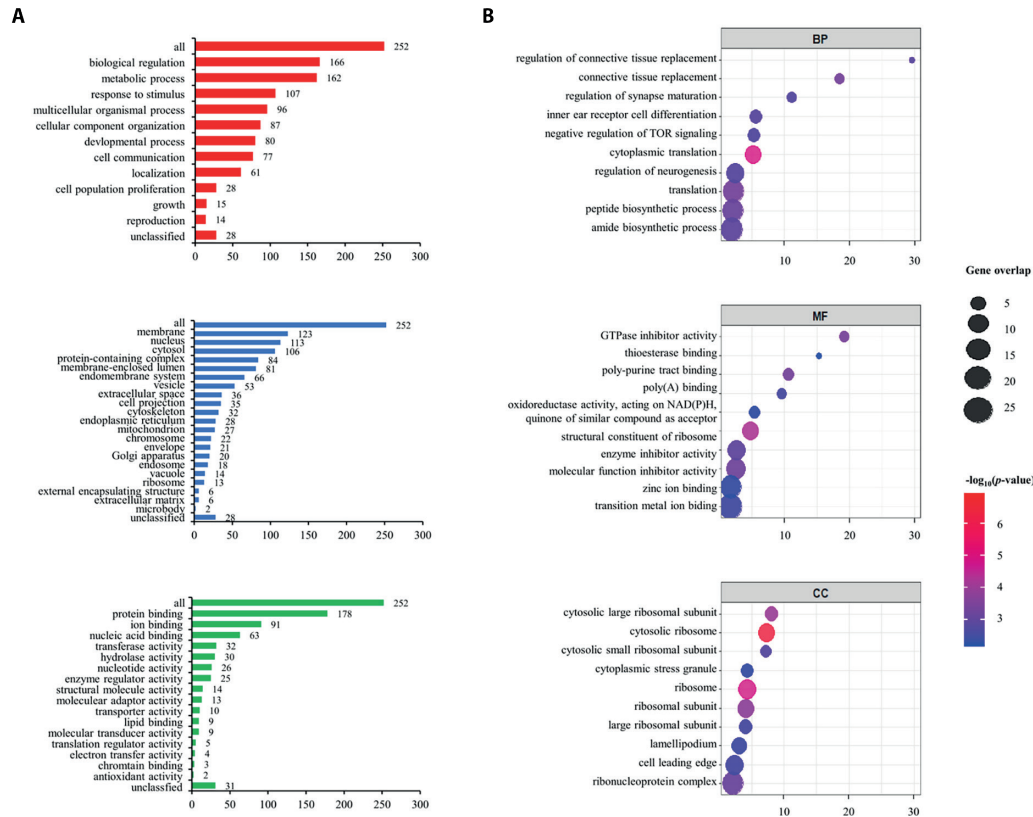


Figure 4. Biological and cellular ontology analysis of the 416 genes associated with fibroblast growth slope. (A) Bar charts showing the GO classification of the candidate genes into three main categories: BP (red), CC (blue), and MF (green). The x-axis indicates the number of genes, and the y-axis represents the GO classification terms. Numbers next to the bars denote the number of genes associated with each category. (B) Dot plots showing the GO enrichment analysis results for BP, MF, and CC. The x-axis represents the enrichment ratio, and the y-axis represents the enriched GO terms. Bubble size reflects the number of overlapping genes, and the color gradient indicates statistical significance [$-\log_{10}(p\text{-value})$], ranging from blue (lower significance) to red (higher significance). *Abbreviations:* GO, Gene Ontology; BP, biological process; MF, molecular function; CC, cellular component.

Table 2. Identification of candidate seven genes correlated with fibroblast growth slope.

Genes	IPF vs Control (DEG)				Correlation with slope	
	$\log_2\text{FC}$	$\log\text{CPM}$	p-value	FDR	r	p-value
PITX1	3.87	6.3	0.004	0.523	0.714	0.047
DHFR	2.73	0.82	0.004	0.523	-0.905	0.002
LOC100129906	2.56	2.15	0.035	0.564	-0.762	0.028
IQGAP2	2.39	0.44	0.031	0.556	0.762	0.028
LOC100134466	1.67	1	0.021	0.543	0.81	0.015
LYPD6	-1.58	1.86	0.046	0.58	-0.833	0.01
ENPP6	-2.22	0.71	0.002	0.523	0.747	0.033

Abbreviations: DEG, differentially expressed gene; FC, fold change; FDR, false Discovery Rate; CPM, counts per million; $\log\text{CPM}$, $\log_2$ -transformed counts per million; r, Spearman's rank correlation coefficient.

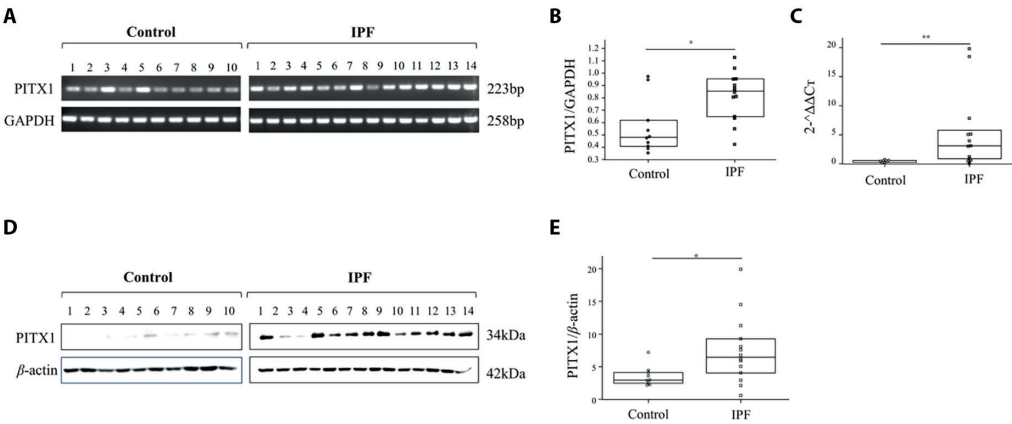


Figure 5. Protein and mRNA levels of PITX1 in lung tissue-derived fibroblasts from 14 IPF patients and 10 controls. (A) RT-PCR and (B) densitometry of PITX1 band intensity normalized to GAPDH. (C) Real-time PCR analysis normalized to β -actin. PITX1 mRNA levels in the IPF group were significantly higher than those in the control group (* $p < 0.05$ and ** $p < 0.005$, respectively). (D) Western blot and (E) densitometry of PITX1 protein band intensity normalized to β -actin (* $p < 0.05$). Data are presented as medians with 25% and 75% quartiles.

Table 3. Clinical profile of study subjects.

Parameters	Control	IPF
No.	30	84
Age (years)	55(35-72)	63.41(32-85)
Sex (male/female)	17/13	51/33
Smokers (NS/ES/SM)	ND	38/24/22
Follow up duration (years)	ND	4.90 (2.35-6.40)
FVC (% pred.)	115(106.2-132.2)	75.5(65.0-84.7)
DLCO (% pred.)	ND	63.0 (0.0-71.7)
BAL total cell count ($\times 10^5/\text{mL}$)	0.41 $\pm$ 0.05	4.81 $\pm$ 0.29
Macrophages (%)	94.95 $\pm$ 9.12	56.01 $\pm$ 24.98
Neutrophils (%)	3.36 $\pm$ 4.92	26.07 $\pm$ 21.31
Eosinophils (%)	0.55 $\pm$ 1.02	3.26 $\pm$ 8.02
Lymphocytes (%)	2.43 $\pm$ 4.52	5.72 $\pm$ 4.71

Abbreviations: IPF, idiopathic pulmonary fibrosis; NS/ES/SM, non-smokers/ex-smokers/smokers; ND, not determined; FVC, forced vital capacity; DLCO, diffusing capacity of the lung for carbon monoxide. Patient characteristics and pulmonary function test results are presented as medians (interquartile ranges) and were compared between the two groups using the Mann–Whitney U test. Bronchoalveolar lavage cell numbers are presented as means $\pm$ standard errors of the mean (SEM) and were compared between the two groups using Student’s t-test. A p-value < 0.05 was considered statistically significant.

that increased fibroblast growth slopes are associated with poorer survival outcomes in IPF. By integrating fibroblast growth kinetics with transcriptome microarray data, we identified a fibroblast growth slope-associated gene signature and further refined this signature to a disease-relevant candidate gene, PITX1. We show

that PITX1 expression is elevated at both the mRNA and protein levels in IPF fibroblasts and that increased PITX1 protein levels in BALFs are associated with better survival outcomes, supporting a complex relationship between fibroblast proliferative behavior, molecular dysregulation, and prognosis in IPF.

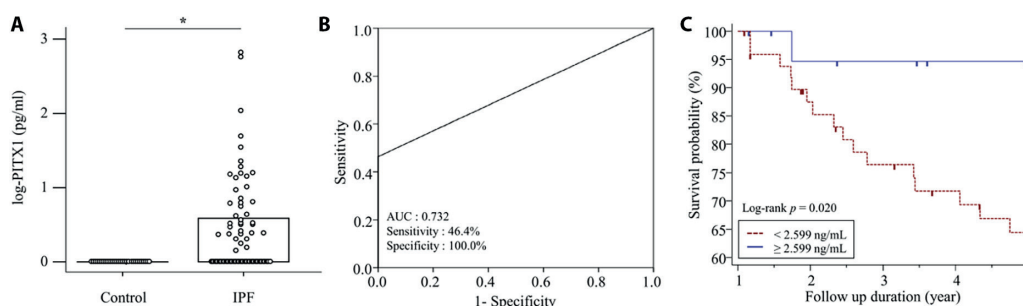


Figure 6. Comparison of PITX1 levels among control and IPF. (A) PITX1 protein concentrations in BALFs. PITX1 protein was detected in 30 controls and 84 IPF patients. Data were presented as median values with 25 and 75% quartiles. Statistical significance was determined compared to the control group (* $p < 0.001$). (B) Receiver operating characteristic (ROC) curve of the PITX1 protein concentration between the IPF and control. A cut-off value 0.53pg/mL had 100% specificity, 46.4% sensitivity between the IPF and control. (C) Kaplan-Meier survival curves of patients with IPF stratified by BAL PITX1 levels ($n = 80$). Patients were divided into a low BAL PITX1 group (< 2.599 ng/mL, red dashed line, $n = 54$) and a high BAL PITX1 group (≥ 2.599 ng/mL, blue solid line, $n = 26$). The high BAL PITX1 group showed significantly better survival than the low BAL PITX1 group (log-rank test, $p = 0.02$; HR, 0.29; 95% CI, 0.102–0.821).

It has been reported that fibroblasts isolated from the lungs of patients with IPF exhibit higher proliferative capacity than those from normal lungs, largely due to altered interleukin-6 (IL-6)/gp130-mediated signaling (27,28). In contrast to normal fibroblasts, in which IL-6 suppresses proliferation, IPF fibroblasts respond to IL-6 with enhanced mitogenic signaling, and this response is heterogeneous among IPF patients, reflecting differences in STAT and MAPK pathway activation (29). In addition, Toll-like receptor 9 (TLR9) expression is increased in IPF fibroblasts and is further enhanced by Th2 cytokines, with fibroblasts from rapid progressor IPF patients showing heightened responsiveness to TLR9 stimulation compared to those from slow progressors (30,31). Collectively, these findings support the concept that heterogeneity in fibroblast proliferative behavior contributes to disease progression in IPF. Consistent with this notion, our study demonstrates that a higher fibroblast growth slope is significantly associated with reduced survival and increased mortality in IPF patients during long-term follow-up, indicating that accelerated fibroblast proliferation is linked to poor prognosis in IPF.

To further investigate the molecular drivers of fibroblast heterogeneity in IPF, we focused on identifying genes whose expression levels directly reflect the cellular phenotype, specifically fibroblast proliferation rate. Among the 15,020 genes analyzed, the expression

levels of 416 genes showed a significant correlation with the fibroblast growth slopes of IPF-derived fibroblast ($p < 0.05$) and their expression patterns across the eight IPF samples are summarized in Supplemental Table 2. Functional enrichment analysis revealed that these growth-correlated genes were strongly associated with connective tissue remodeling programs, including *regulation of connective tissue replacement* and *connective tissue replacement*, which encompasses key fibrotic regulators such as ROCK1, HIF1A, and PPP3CA. These findings suggest that fibroblast proliferative capacity is intrinsically linked to their ability to remodel extracellular matrix in IPF. In addition, a prominent enrichment of translation-related biological processes, including *cytoplasmic translation* and *peptide biosynthetic process*, was observed, indicating that highly proliferative IPF fibroblasts possess enhanced protein synthesis machinery. Collectively, these results define a transcriptomic signature of highly proliferative IPF fibroblasts characterized by enhanced translational activity and connective tissue remodeling potential, which may contribute to fibroblast heterogeneity and fibrotic progression in IPF (Figure 4, Supplemental Table 3).

Of the 416 genes correlated with fibroblast proliferation and the 1,274 genes differentially expressed between IPF and control fibroblasts (Supplemental Table 4), PITX1 was the only gene shared between the two datasets (Table 2). PITX1 encodes a homeobox

transcription factor that has been implicated in the regulation of cellular development, differentiation, and transcriptional control (32–34). Genetic and epigenetic dysregulation of *PITX1* has been associated with various pathological conditions, including lung adenocarcinoma (35), idiopathic scoliosis (36), head and neck squamous cell carcinomas (37,38) and progression and prognosis of cutaneous malignant melanoma (39). In the context of fibrotic lung disease, these transcriptional functions suggest that *PITX1* may be associated with fibroblast responses to chronic tissue injury and remodeling. However, *PITX1* should be interpreted as a growth slope–associated candidate gene rather than as a confirmed mechanistic regulator of fibroblast proliferation or disease progression. Although *PITX1* expression was increased in IPF fibroblasts and positively correlated with fibroblast proliferation rates, higher *PITX1* expression was paradoxically associated with improved survival outcomes, indicating that *PITX1* is unlikely to act as a direct driver of disease progression in IPF. Rather, the present findings raise the possibility that *PITX1* induction may represent a compensatory or context-dependent transcriptional response.

Indeed, previous studies have demonstrated that *PITX1* overexpression suppresses cell proliferation and cell cycle progression in multiple cellular contexts through regulation of cell cycle–associated pathways and tumor suppressive mechanisms (39–42). In addition, *PITX1* has been implicated in the modulation of cellular senescence and stress-responsive transcriptional programs (39,43,44). Given that IPF is characterized by aberrant wound healing with excessive fibroblast proliferation, resistance to apoptosis, and altered senescence (45–47), the upregulation of *PITX1* in IPF fibroblasts may reflect an intrinsic regulatory response associated with fibroblast activation. However, whether *PITX1* directly modulates fibroblast overactivation or primarily reflects disease-related fibroblast heterogeneity remains to be determined. Future studies employing functional perturbation of *PITX1* will be required to elucidate its precise role in IPF fibroblast biology and its potential contribution to the clinical heterogeneity of IPF.

IPF and progressive fibrotic ILDs are characterized by substantial heterogeneity in disease trajectory, functional decline, radiological progression, and

response to antifibrotic therapy. In real-world settings, treatment outcomes may also be influenced by treatment timing, longitudinal management strategies, therapeutic adherence, and dosing patterns (48,49). In this context, fibroblast growth behavior and associated molecular signatures may provide a biological framework for future individualized risk stratification in fibrotic lung diseases.

Although faster fibroblast proliferation was associated with poorer prognosis in IPF, neither fibroblast growth slope nor *PITX1* mRNA expression correlated with pulmonary function parameters, which represents an important limitation of this study. In addition, transcriptome analyses were performed in a limited subset of IPF fibroblast samples with matched fibroblast growth slope data, resulting in reduced statistical power. Therefore, the correlation-based transcriptomic findings should be interpreted as exploratory and hypothesis-generating. Further validation in larger, independent cohorts is required to confirm the robustness of the growth slope–associated transcriptomic signature. Another limitation is that control fibroblasts were derived from lung cancer tissue, which may not fully represent normal fibroblast biology. Finally, the discordance between *PITX1* mRNA expression in fibroblasts and *PITX1* protein levels in BALFs, which showed different prognostic associations, warrants further validation. Although BAL *PITX1* levels were significantly higher in patients with IPF and showed prognostic relevance, the relatively low sensitivity observed in ROC analysis indicates that its clinical utility as a standalone biomarker remains limited. In addition, BAL-based measurements may be affected by procedural variability, including sampling site, lavage volume, recovery rate, and sample processing. These BAL-based findings require validation in larger prospective cohorts before clinical application can be considered.

Conclusions

Enhanced fibroblast proliferative capacity was associated with poor survival outcomes in patients with idiopathic pulmonary fibrosis. In addition, *PITX1* emerged as a fibroblast growth slope–associated

gene that reflects fibroblast heterogeneity and shows prognostic relevance depending on biological compartment. These findings suggest that assessment of fibroblast proliferative behavior and PITX1 expression may provide insight into disease progression in IPF, although further validation and functional studies are required.

Ethic Approval: samples were obtained after the approval of the study protocol by the Ethics Committee of Soonchunhyang University Hospital (SCHCA-IRB-2018-10-034 and 201910-BR-058). Informed written consents for study participation and sample donation were obtained from each subject

Authors' Contribution: Conceptualization, M.K.K., W.S.P., and J.-U.L.; Data curation, M.K.K., S.-L.P. and E.J.S.; Formal analysis, S.-L.P. and E.J.S.; Funding acquisition, W.S.P. and J.-U.L.; Investigation, M.K.K. and J.-U.L.; Methodology, S.-L.P. and E.J.S.; Project administration, S.W.P.; Resources, W.S.P. and J.-U.L.; Software, M.K.K. and J.-U.L.; Supervision, W.S.P. and J.-U.L.; Validation, M.K.K., S.-L.P. and E.J.S.; Visualization, M.K.K. and J.-U.L.; Writing—original draft preparation, M.K.K. and J.-U.L.; Writing—review and editing, All authors. All authors have read and agreed to the published version of the manuscript.

Conflict of Interest: The authors declare that they have no competing interests.

Declaration on the Use of AI: During the preparation of this work the authors used ChatGPT in order to English language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data Availability Statement: Data is freely available.

Acknowledgements: The authors thank the participants of the study as well as all the study staff for their contributions to the study.

Funding: This research was supported by the National Research Foundation of Korea (NRF) through the Basic Science Research Program funded by the Ministry of Education (RS-2025-25419099) and a grant funded by the Korea government (MSIT) (RS-2023-00274884), as well as the

Soonchunhyang University Research Fund. The authors thank the participants of the study as well as all the study staff for their contributions to the study.

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Appendix – Supplementary Tables

Supplementary Tables (S1-S4) can be found online

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